



VS Code Copilot 1.121–1.132

Portable Infrastructure, Open Models, and a Closed Review Loop

Bring any model — including local Ollama — with no GitHub sign-in required

"How does VS Code Copilot evolve from a window-bound chat to portable agent infrastructure?"

Releases 1.121–1.132 change how agents are deployed, which models they use, and how review happens.

Which capability is your team ready to act on tomorrow?

Developer

Which new models and local options can I use without changing my GitHub account?



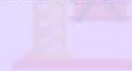
Tech Lead

How do parallel sessions, worktrees, and remote hosts change how we assign agent work?



Platform Engineer

What controls exist for model gateways, MCP allowlists, and org-wide policy enforcement?



Open the Agents window

Dedicated agent-first surface — still missed by teams stuck in side chat

BYOK GA + Ollama

No GitHub sign-in needed for chat, tools, and MCP; local models auto-discovered

Vision paste + browser loop

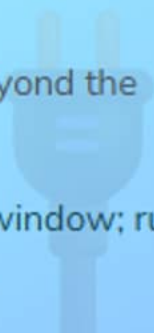
Attach Excel/screenshots as images; GA browser tools with element comments

PART 1

Agent Infrastructure

AHP + SDK: sessions persist beyond the window

Decouple the session from the window; run agents on owned remote hosts

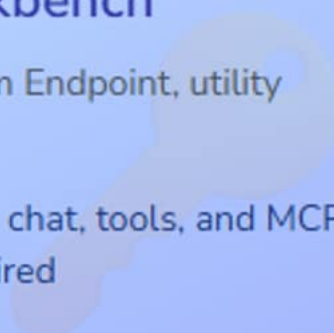


PART 2

Open Model Workbench

BYOK GA, Ollama, Custom Endpoint, utility models

Any compatible model for chat, tools, and MCP — no GitHub sign-in required

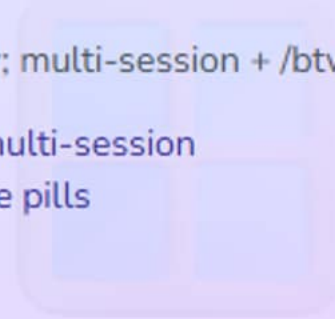


PART 3

Parallel Agent Work

Open the Agents window; multi-session + /btw

Awareness first — then multi-session workspace, side chats, live pills

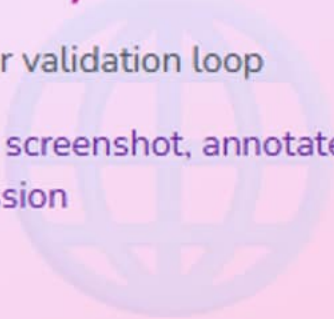


PART 4

Closed-Loop Delivery

Vision paste + GA browser validation loop

Attach evidence, emulate, screenshot, annotate — without leaving the session



Click any section to jump directly there

Agent Infrastructure

The Copilot SDK and Agent Host Protocol detach the session from the window so agents run on infrastructure you own



Agent Host Protocol

Open spec: host owns state, clients sync and disconnect



Remote Machine Host

SSH or dev tunnel; session persists after VS Code closes



Copilot SDK Runtime

Copilot, Claude, Codex harnesses in one dedicated process

Preview: The session is no longer the window
↳ session persists on remote infra → reconnect on completion

Session Is No Longer the Window

🖥️ Before AHP

Close VS Code → agent stops

Session lifetime = window lifetime

Context is lost on disconnect

One active session per open window

Long autonomous tasks compete with active work

🔧 With AHP (Preview)

Host persists after VS Code closes

Reconnect to a running session any time

Multi-client: same session, multiple windows

Long tasks run on infrastructure your team owns

Progress, tool outputs, and memory survive disconnect

Connect a Remote Agent Host in Three Steps

1

Open Agents Window

Go to the Remote tab inside the Agents window in VS Code stable preview

No separate tooling required

2

Connect via SSH or Tunnel

Use an existing `~/.ssh/config` entry, `user@host` string, or a running dev tunnel

Any existing SSH config works

3

Host Starts on Remote

VS Code installs its CLI server and starts the agent host on the target machine

Host persists after VS Code closes

v1.132: connect from multiple windows

PART 2

Open Model Workbench

BYOK is GA and needs no GitHub sign-in — any compatible provider drives chat, tools, and MCP



BYOK Without Sign-In

Any provider key — GA, no GitHub account needed



Local via Ollama

Pull models locally; zero network requests, air-gap ready



Custom + Utility

Stable gateway endpoint + per-task model tuning

PRIMARY DEMO: model picker → Manage Models... → add provider
↳ models appear instantly – billing and rate limits via provider

BYOK is GA: No GitHub Sign-In Required



Cloud Providers

Anthropic, Azure OpenAI, Gemini, OpenAI, OpenRouter — add key in Manage Models...

Billing and rate limits via provider

No Copilot quota consumed



Local via Ollama

VS Code auto-discovers pulled Ollama models — zero network requests leave the machine

Air-gap and data-residency safe

Full chat, tools, and MCP support



One Firm Boundary

Inline completions and NES still require GitHub sign-in — BYOK covers chat and agents only

No BYOK path for ghost-text completions

Plan inline suggestions separately

Local Models via Ollama: Zero Network Requests

terminal

```
# 1. Install Ollama (ollama.com)
# 2. Pull models locally
ollama pull llama3.2
ollama pull codestral

# 3. In VS Code:
# model picker → Manage Models... → Ollama
# VS Code discovers pulled models automatically
```

Auto-Discovery

VS Code finds all locally pulled Ollama models without manual registration

Air-Gap Ready

Zero network requests — data residency guaranteed for sensitive codebases

Full Feature Parity

All chat features, tool execution, and MCP work with local Ollama models

Stable Custom Endpoint and Utility Model Tuning

settings.json

```
{  
  // Route all VS Code Copilot chat through  
  // your own gateway – Chat: Manage Language Models  
  // Add → Custom Endpoint → choose API family  
  
  // Split utility tasks to a cheaper model  
  "chat.utilityModel": "[configured-model-id]",  
  "chat.utilitySmallModel": "[fast-local-model-id]"  
}
```

Custom Endpoint (Stable)

Any Chat Completions, Responses, or Messages
API gateway — stable since v1.122

Utility Model Split

Frontier model for primary work; fast local model
for titles, summaries, and commits

Cost and Compliance

Route all chat traffic through your internal
gateway for logging and cost control

Model Selection: Four Paths for Four Needs



Interactive Chat

Frontier model via Copilot or BYOK provider key



Large Codebase Analysis

BYOK provider with 1M-token context + direct API access



Utility Sub-Tasks

Fast local or low-cost model for titles, summaries, commit messages



Sensitive or Air-Gap

Ollama local model — zero data leaves the machine

Parallel Agent Work

First open the Agents window — then multi-session work, side chats, and live activity become obvious



Open It

Title bar, Command Palette, or code --agents



Multi-Session Workspace

Isolated worktrees, tools, and model config per session



/btw + Live Pills

Lateral questions and at-a-glance activity without polling

Most teams still only use side chat

↳ Open the Agents window before teaching parallel features

Open the Agents Window First



Title Bar

Select Open in Agents in the VS Code title bar — always one click away



Command Palette

Chat: Open Agents Window via Ctrl+Shift+P / ⌘⇧P



CLI

`code --agents` from the terminal — scriptable entry point



Browser / Welcome

Welcome-page link or insiders.vscode.dev/agents

Dedicated VS Code window beside the editor — same sessions as side Chat, agent-first layout

Agents Window: From Side Chat to Workspace

Side Chat Only (common habit)

One conversation buried in the editor sidebar

No multi-workspace sessions list

Hard to track concurrent agent work

Changes and files live in other panels

Agents Window (Preview workspace)

Dedicated window: sessions across workspaces

Chat + Changes + Files for the active session

Multiple sessions side-by-side with isolated worktrees

/btw side chats and live activity pills

/btw Side Chats and Live Activity Pills

/btw Side Chat (GA v1.132)

Opens without pausing the active turn

Main agent keeps executing while you type

Shares context and prompt cache with main chat

Captures observations without interrupting the task

/btw should we use the builder pattern here?

Ask mid-tool-chain; main turn keeps running

Live Activity Pills (GA v1.132)

Changes, Previews, Subagents, Browsers

All visible at a glance in the Agents window

Navigate directly to an active subagent or browser

Pills clear when a turn completes — no noise accumulation

Replaces manual polling for Autopilot sessions

No need to check the terminal and come back

Closed-Loop Delivery

Vision paste for everyday evidence — then GA browser tools close the build-
validate-review loop



Copilot Vision (GA)

Paste Excel, mockups, PDFs — attach evidence, not walls of text



GA Browser Tools

Navigation, screenshot, interaction — on by default



Element Comments

Select elements in v1.132, anchor a comment per element

CLIMAX DEMO: `implement` → `emulate` → `screenshot` → `annotate elements` → `re-validate`

↳ Vision paste for human evidence; browser tools for agent validation

Paste Evidence Instead of Walls of Text

Copilot Vision (GA v1.128)

Paste, drag-drop, or attach images and PDFs

Ask, plan, and agent modes

Available on Free, Pro, Pro+, Business, and Enterprise

Excel range → image attachment

Clipboard image data attaches as vision context

Same path for mockups, error dialogs, whiteboards

Browser Add-to-Chat

Add Screenshot to Chat

Capture the current viewport as an image

Add Element to Chat

Selected elements with styles and screenshots

Add Console Logs to Chat

Runtime output without copy-paste sprawl

You already have the page — attach it and ask

Browser Validation: Before vs After GA Tools

Closing the loop without leaving the session

✗ Manual Browser Validation

- 1 Code the UI change
Edit in the editor
- 2 Open browser manually
Switch to a separate browser tab
- 3 Resize manually for mobile
Approximate the device viewport
- 4 Annotate in external tool
Screenshot, paste, mark up separately
- 5 Copy feedback back into chat
Describe the issues in words



3-5 context switches per iteration

✓ With GA Browser Tools

- 1 Code the UI change
Edit in the editor
- 2 Agent opens localhost page
Browser opens in the integrated panel
- 3 Agent emulates device preset
iPhone 15, Galaxy S24, or custom viewport
- 4 Screenshot attaches to chat turn
Validation evidence is part of the conversation
- 5 Select elements, anchor a comment each
Precise per-element feedback (v1.132)



Full loop inside one session

3-5 context switches per iteration → 0

The Closing Demo: Emulate, Screenshot, Annotate

Implement → emulate → screenshot → element comment → re-validate, all in one session

```
$ agent
> "Implement the mobile login form and validate it on iPhone 15"
Copilot agent – browser tools active:
  Opening http://localhost:3000/login ...
  Emulating iPhone 15 (390x844 viewport) ...

✓ Screenshot attached to this turn: login-mobile.png
> "[element comment on .btn-submit] Tap area is 28px – needs 44x44px min for accessibility"

✓ Applied: min-height: 44px; padding: 12px on .btn-submit
✓ Screenshot attached: login-mobile-v2.png – tap area confirmed
```

implement → emulate → screenshot → annotate elements → re-validate: all in one session

Where the Loop Is Heading (Preview)



In-Window Diff Review

Apply, revert, or cherry-pick per file; inline comments the agent acts on in later turns

Preview in Agents window



CI Feedback in Session

Failed check details surface in the session — read, fix, and re-run without tab-switching

Preview: CI response in-window



PR Comment Response

PR review comments surface on the branch; respond, apply, or mark addressed in-window

Preview: PR feedback in-session

From Window-Bound Sessions to Portable Agent Infrastructure

Before

Agents only used from the side Chat view

One provider; GitHub sign-in required for chat and tools

Paste walls of Excel/text instead of visual evidence

Manual browser validation with context switches per iteration

After

Agents window open as the multi-session agent surface

Any BYOK provider or local Ollama model — no GitHub sign-in

Vision paste for Excel/mockups; browser Add Screenshot to Chat

GA browser loop with device emulation and element comments

Open

Agents window — title bar, palette, or code `--agents`

GA

Vision paste + browser tools with device emulation

0

GitHub sign-in required for chat, tools, and MCP

✓ What You Can Do Today

Today

- ❑ Open the Agents window (title bar, Command Palette, or code `--agents`)
- ❑ Paste an Excel range or screenshot into Chat — Vision attaches it as an image
- ❑ Open model picker → Manage Models... → add a BYOK or Ollama provider

This Week

- ❑ Try `/btw` during a long agent turn; watch live activity pills
- ❑ Use browser Add Screenshot / Element to Chat on a localhost page
- ❑ Run a UI task and validate it with GA browser device emulation

This Month

- ❑ Evaluate Custom Endpoint for team-wide cost and compliance routing
- ❑ Run parallel Agents window sessions for independent concurrent tasks
- ❑ Review session diffs in-window and respond to CI feedback in-session

🔑 Key Takeaway

Open the Agents window, bring any model, and close the loop with vision + browser evidence.

 Official Documentation[VS Code release notes: August 5, 2026 \(v1.132\)](#)

Element comments, /btw, live pills, multi-window agent sessions

[Use the Agents window \(Preview\)](#)

How to open it, multi-workspace sessions, Changes/Files panels

[VS Code release notes: July 8, 2026 \(v1.128\)](#)

Copilot Vision GA — paste images and PDFs into Chat

[Copilot vision is generally available](#)

Supported formats and plan availability for image/PDF attachments

[Add context to chat](#)

Vision attachments and browser Add Screenshot / Element to Chat

[GitHub Copilot in VS Code: July 2026 releases](#)

/btw side chats, live activity pills, and Agents window workspace updates

 Related Content[Copilot SDK Talk](#)

Deep dive into the Copilot SDK harness and Agent Host Protocol

[MCP Apps Talk](#)

Connecting MCP servers to VS Code Copilot sessions and workflows

[Upcoming September 2026 model deprecations in GitHub Copilot](#)

Context for why this talk uses durable frontier-model guidance instead of named catalog entries



VS Code Copilot 1.121–1.132

Portable Agent Infrastructure, Open Models, and a Closed Review Loop

Agents Window

Open it — multi-session agent surface most teams still skip

BYOK GA

Any model, no GitHub sign-in — chat, tools, and MCP

Vision GA

Paste Excel/screenshots/PDFs instead of walls of text

Browser GA

Emulate, screenshot, element comments — no context switch

Which changes your team first — opening the Agents window, model choice, or vision + browser evidence?